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APOLLO AGRIVERSE — INTELLIGENT SOIL DIVISION

Project Abstract & Outline

DEVELOPMENT OF A PHYSICS-INFORMED MACHINE LEARNING FRAMEWORK FOR DECOUPLING HYDROGEL HYDRATION FROM BULK SOIL MOISTURE IN INTELLIGENT SOIL SYSTEMS

Course Title: Engineering Internship Project
Course No.: INT-2026-02

PROJECT EXECUTION DETAILS

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Degree Program: Diploma in Computer Engineering

Research Area: Applied Machine Learning, Computational Physics & AgriTech Systems

PROJECT CARRIED OUT AT:

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1. Broad Area of Work

The broad area of this project lies at the intersection of Computational Materials Science, Applied Artificial Intelligence, and Precision AgriTech, developed in support of the Apollo AgriVerse Intelligent Soil initiative at Chatake Innoworks Pvt. Ltd. The project focuses specifically on physics-informed machine learning: the practice of constraining neural network training with governing physical equations so that model outputs remain scientifically consistent even when field data is sparse or noisy. The core technological domains involved include:

- **Multiphysics Simulation:** Coupling of unsaturated soil-water flow physics (the Richards equation) with polymer swelling thermodynamics (Flory-Rehner theory) to model how stimuli-responsive hydrogels interact with surrounding soil.
- **Physics-Informed Neural Networks (PINNs):** Embedding differential-equation residuals directly into the neural network loss function so predictions of soil moisture and hydrogel hydration remain physically admissible.
- **Sensor Signal Decoupling:** Engineering software-level correction algorithms to separate hydrogel-induced dielectric and mechanical interference from true bulk soil moisture readings captured by flexible capacitive geotextile sensors.
- **Edge-AI & Digital Twin Software:** Designing lightweight inference pipelines suitable for microcontroller deployment, feeding into a cloud-based digital twin dashboard for irrigation decision support.

2. Background

Water scarcity, soil degradation, and climate variability present acute challenges to Indian and global agriculture. Passive soil amendments such as superabsorbent hydrogels reduce irrigation demand effectively: polyacrylamide-based hydrogels can absorb several hundred times their dry weight in water and release it gradually to plant root zones, with field trials reporting soil moisture increases of roughly 25% and meaningful reductions in nitrogen leaching under water-stressed conditions (Krasnopeeva et al., 2022; PMC12027221, 2025; PMC9824677, 2023). Apollo AgriVerse's Intelligent Soil concept extends this using thermoresponsive and pH/ionic-triggered polymer networks that autonomously swell and release water and nutrients in response to local soil conditions — effectively creating "soil that thinks" without embedded electronics.

In parallel, flexible geotextile-integrated capacitive sensors have emerged as a low-cost way to monitor field-scale soil moisture, using interdigitated electrodes printed on flexible substrates whose capacitance shifts with surrounding dielectric permittivity (ScienceDirect S0924424724002905, 2024; Nature Communications, 2026). A critical instrumentation problem arises, however, when the two technologies are combined: a hydrogel node swelling next to a sensor trace sharply raises the local dielectric permittivity and mechanically strains the fabric, causing the sensor to register an artificially saturated "wet zone" instead of true bulk soil moisture — rendering the data unreliable for irrigation scheduling unless corrected.

Traditional numerical solvers for the governing water-flow physics (the Richards equation) are accurate but computationally expensive for real-time, in-field inference (ERDC PINN Soil Moisture study, 2023). Physics-Informed Neural Networks (PINNs) can approximate Richards/Richardson-Richards equation solutions far faster than finite-difference or finite-element solvers while remaining physically consistent, and can be used inversely to recover soil hydraulic properties directly from noisy moisture measurements (Bandai & Ghezzehei, 2022; Hess.copernicus.org, 2022; Nature Scientific Reports, 2025). No existing published framework, however, couples this Richards-equation PINN methodology with Flory-Rehner polymer swelling thermodynamics specifically to decouple hydrogel hydration signal from bulk soil signal at a flexible geotextile sensor interface — the gap this project addresses.

3. Objectives

The primary objectives of this internship project are as follows:

- **Multiphysics Model Definition:** Formulate the coupled governing equations — the Richards equation for bulk soil-water flow and Flory-Rehner theory for hydrogel swelling — that describe the hydrogel-soil-sensor interface.
- **Synthetic Dataset Generation:** Produce labeled training and validation data representing sensor readings under varying hydrogel proximity, soil texture, and moisture conditions, using finite-element or finite-difference reference solutions.
- **PINN Model Development:** Train a Physics-Informed Neural Network whose loss function is constrained by both governing equations, capable of predicting true bulk soil moisture independent of local hydrogel interference.

- **Signal Decoupling & Validation:** Quantify the accuracy improvement of the physics-informed approach over a standard (non-physics-constrained) neural network baseline using error metrics such as RMSE and R^2.
- **Lightweight Deployment Prototype:** Package a simplified, computationally efficient version of the trained model into a Python-based Edge-AI inference script suitable for future microcontroller deployment.

4. Scope of Work

The scope of this project is limited to the design, simulation, and software validation of the physics-informed decoupling model. The student will independently architect the Python-based simulation and machine learning pipeline, generate synthetic multiphysics datasets, and produce a validated PINN prototype along with supporting visual and statistical analysis.

Limitations: For this 4-week timeline, the project will not include physical fabrication or laboratory testing of hydrogel formulations or geotextile sensor hardware, nor will it include live field deployment or streaming IoT telemetry. The work is confined to the computational and data-science layer — building and validating the simulation and inference software that will later interface with Apollo AgriVerse's physical hydrogel and sensor hardware once fabricated.

5. Plan of Work

Given the 4-week timeframe for this internship project, the execution is structured into the following operational phases:

Execution Phase	Duration	Milestones & Deliverables
Phase 1: Architecture & Design	Week 1	Literature review of Richards-equation PINNs and Flory-Rehner hydrogel theory; environment setup (Python, PyTorch, virtual environments); drafting of the Project Abstract Document.
Phase 2: Data & Physics Modeling	Week 2	Formulation of the coupled governing equations; generation of synthetic multiphysics datasets representing hydrogel-adjacent and bulk-soil sensor readings.
Phase 3: PINN Development & Testing	Week 3	Construction and training of the physics-informed neural network; benchmarking against a standard neural network baseline; error and accuracy analysis.

Execution Phase	Duration	Milestones & Deliverables
Phase 4: Capstone Assembly	Week 4	Packaging of a lightweight Edge-AI inference prototype; final report generation; publishing of the repository to GitHub.

6. Literature References

The following academic and technical resources establish the foundational methodology used in the design of this simulation and modeling framework:

1. Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational Physics, 378, 686–707.

2. Richards, L. A. (1931). Capillary conduction of liquids through porous mediums. Physics, 1, 318–333.

3. Bandai, T., & Ghezzehei, T. A. (2022). Forward and inverse modeling of water flow in unsaturated soils with discontinuous hydraulic conductivities using physics-informed neural networks with domain decomposition. Hydrology and Earth System Sciences, 26, 4469–4495.

4. Bandai, T., & Ghezzehei, T. A. (2024). Physics-informed neural networks for the Richardson-Richards equation: Estimation of constitutive relationships and soil water flux density from volumetric water content measurements. Water Resources Research (related ESS Open Archive preprint, 2022).

5. Solving the Richards infiltration equation by coupling physics-informed neural networks with Hydrus-1D. (2025). Scientific Reports / PMC12120032.

6. Krasnopeeva, E. L., Panova, G. G., & Yakimansky, A. V. (2022). Agricultural Applications of Superabsorbent Polymer Hydrogels. International Journal of Molecular Sciences, 23(23), 15134.

7. Hydrogel Performance in Boosting Plant Resilience to Water Stress — A Review. (2025). PMC12027221.

8. Superabsorbent Polymers as a Soil Amendment for Increasing Agriculture Production with Reducing Water Losses under Water Stress Condition. (2023). PMC9824677.

9. Highly sensitive screen-printed soil moisture sensor array as green solutions for sustainable precision agriculture. (2024). Sensors and Actuators B: Chemical, ScienceDirect.

10. Cellulose-based sensors for decentralized monitoring in precision agriculture. (2026). Nature Communications.

11. Flory, P. J., & Rehner, J. (1943). Statistical mechanics of cross-linked polymer networks I: Rubber elasticity. Journal of Chemical Physics, 11, 512–520.

12. Chatake Innoworks Pvt. Ltd. (2026). Apollo AgriVerse: Intelligent Soil Proposal — Materials, Simulation & Deployment Blueprint. Solapur, Maharashtra.

7. Particulars of the Supervisor

Name	Mr. Akash Shivdas Chatake
Qualification	MTech in AI ML
Designation	Founder of Chatake Innoworks Pvt. Ltd. & Research Leader
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8. Remarks of the Supervisor

(To be filled by the supervisor upon technical review of the proposed project abstract. Remarks regarding approval, scope adjustments, and engineering standards will be documented here.)

Date of Review	**Signature of Student**	**Signature of Supervisor**